ALGEBRA FOUNDATIONS

Distributive law

Multiply every term inside brackets by the outside number. For example, 3(x + 2) = 3x + 6. The expression 3x + 2 is not equivalent because the constant 2 must also be multiplied by 3.

Like terms

Like terms have the same variable and power. Combine their coefficients. For example, 4x + 3x = 7x. You cannot combine 4x + 3 into 7x.

Revision: Expand brackets before combining like terms. Check an expansion by substituting x = 1 into both expressions.

LINEAR EQUATIONS

An equation states that two expressions are equal. Keep it balanced: perform the same operation on both sides.

To solve 2x + 6 = 14, subtract 6 from both sides: 2x = 8. Then divide both sides by 2: x = 4. Verify by substitution: 2(4) + 6 = 14.

Fractions

To add fractions with different denominators, use a common denominator. For example, 1/2 + 1/3 = 3/6 + 2/6 = 5/6. Adding the numerators and denominators directly gives an incorrect result.

Revision: Check your answer in the original equation. Find a common denominator before adding fractions.